Architecture

Group 9 - Inside No.9:

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We created our architecture diagrams in PlantUML, using their website's [1] guidance for class diagrams. [2]

Our team brainstormed what the main objects of our system would be, to identify the classes we would need. [Classes Identified]

We discussed and sketched the basic relationship hierarchy of our identified classes. [Class Hierarchy Sketch]
This sketch was formalised into a UML class diagram:

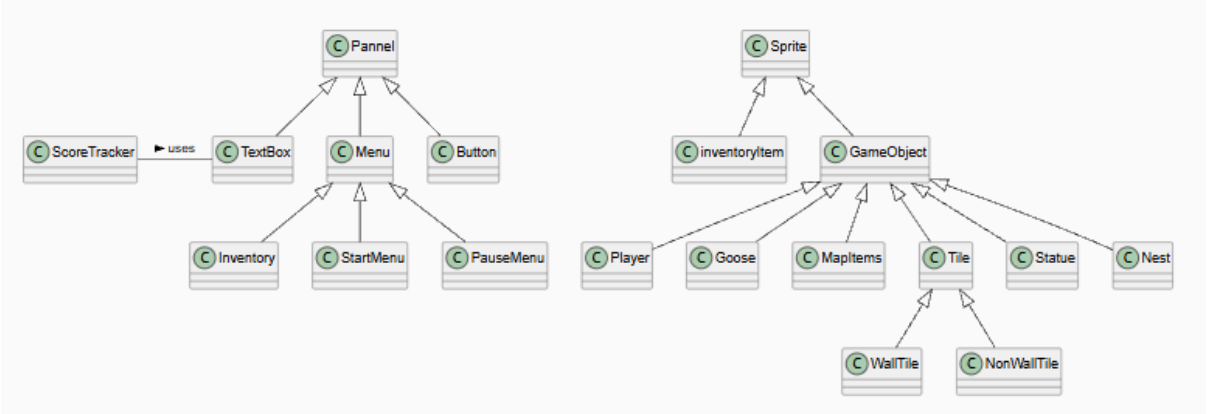


Fig. 1: UML class relationship diagram. Intended to show basic relationships and hierarchies.

By referring back to our requirements to determine what objects would need to 'have', 'do', and how objects would need to interact, we elicited and sketched the attributes, methods and some of the collaboration associations of our identified classes.This was sketched over the top of Fig. 1 [Attributes, Methods & Collaborations]

We formalised this into a more implementation-level UML class diagram [UML Class Diagram 2]

After starting the implementation we wanted to add some new classes: [UML Class Diagram 3]

- LongBoiGoose: extends Goose.
    - We considered Goose extending LongBoi, because Goose would have all the attributes and methods of LongBoi, and then some. Specifically, LongBoi wouldn't have "Distractible"/"Shoo" attribute(s)/method(s), aligning with UR_Geese_Variations. However, we thought it would be confusing to have the main goose class inherit from a unique variant. This explains the somewhat unusual architectural decision to have a child class that doesn't inherit everything from its parent.
- IntractableObject: super class for Statue & Nest.
    - This was refactored to make the code more modular. QR_Maintainabililty.
- Hotbar.

Key Considerations:

(guided by Sommerville's 'Software Engineering' [3])

- Choose the most appropriate structure (e.g client-server, or layered).
- Decide on strategy for decomposing components into subcomponents.
- Develop a model of the control relationships between parts of the system. Also, make decisions about how the execution of components is controlled.

These should depend on the non-functional requirements.

When making project architecture decisions, we tended to lean towards a smaller number of larger components, which meant less communication between components: Beneficial for performance. This suits our QR_Timing_Display and QR_Refresh-Display and therefore our UR_Audiance requirements.

- Security and safety are less relevant requirements: Meaning layered structure is less necessary.
- Availability: This is an important factor to satisfy UR_Accessibilty and UR_OSes. Though, little is needed to change in our architecture to fulfill these requirements.
- Maintainability: Another important consideration, especially as another team will take over this project.

Project evolutions driven by team decisions

Pixel art format: Blow up vs Prefab

Decided on Blow up, implemented using a function of Screen. This aligns our system with our QR_Screen-Size and UR_Audiance requirements, because the system would automatically scale its UI elements, improving legibility for the players.

We decided to change selecting an item from the inventory being controlled by the mouse scroll wheel, to controlled by number keys. The benefit of this was reducing the scope of our project away from harder to implement versions of a similar system, without negatively affecting the player's experience. FR_Inventory_Scroll and FR_Inventory_Highlight were made redundant by FR_Item_Use.

Game loads in its initial state, immediately freezes. Then activate the start menu.

Inventory implemented as a stack with a fixed size.

TMX collision map vs separate painted tiles:

TMX provides more flexibility in changing the look of the world and adding decorations without accidentally breaking our collision layout.

However it requires drawing / managing two parallel layers (the visual map and the collision map).

Collision handled by Player and Goose. Not using listeners. Advantages of not using / using fewer listeners: Fewer classes, limit scope. Remove 'middle men' classes to reduce instructions executed by the system; improves performance - QR_Timing_Display, QR_Refresh-Display, and UR_Audiance.

Change Duck Statue to extend NonWallTile.

New public abstract InteractibleObject class, extends GameObject. Statute and Nest extend this, instead of GameObject.

Behavioural flow for hidden events

Statue event

- Player moves to long-boi statue,
- Press ‘E’ to interact with statue,
- First click → Score increments by 25, events increments by 1
- Second click → Score increments by 25, events increments by 1,
- Third click → Score does not increment, events does not increment and a goose spawns,
- Goose chases player.

Nest event

- Player starts with no egg items in inventory.
- Player moves to within the interaction range of an egg map item.
- System prompts the player to interact using the ‘E’ key.
- Player presses ‘E’ to pick up the egg.
- Map item is removed.
- Egg item is added to inventory. Egg item count in inventory is 1.
- Players moves to within the interaction range of the ‘nest’.
- Player is prompted to interact with ‘E’.
- Player presses ‘E’.
- System increases the player's score value.
- Inventory egg item count is reduced from 1 to 0. Egg item removed from inventory
- Player picks up two other egg map items. A single inventory slot has an egg item with count 2.
- Player interacts with the statue.
- System increases the player's score and reduces egg item count from 2 to 1.
- Player interacts with the statue.
- System creates a new goose at the same location as the statue and reduces egg item count from 1 to 0, removing the egg item from the inventory.

References

[1] PlantUML [Online]. Available: https://plantuml.com

[2] Class Diagrams, PlantUML [Online]. Available: https://plantuml.com/en-dark/class-diagram

[3] I. Somerville. “Architectural Design” In Software Engineering, 10th Global ed. London, England: Pearson, 2015, pp. 168-173. [Online]. Available: https://eu.alma.exlibrisgroup.com/leganto/public/44YORK_INST/citation/67444944630001381?auth=SAML